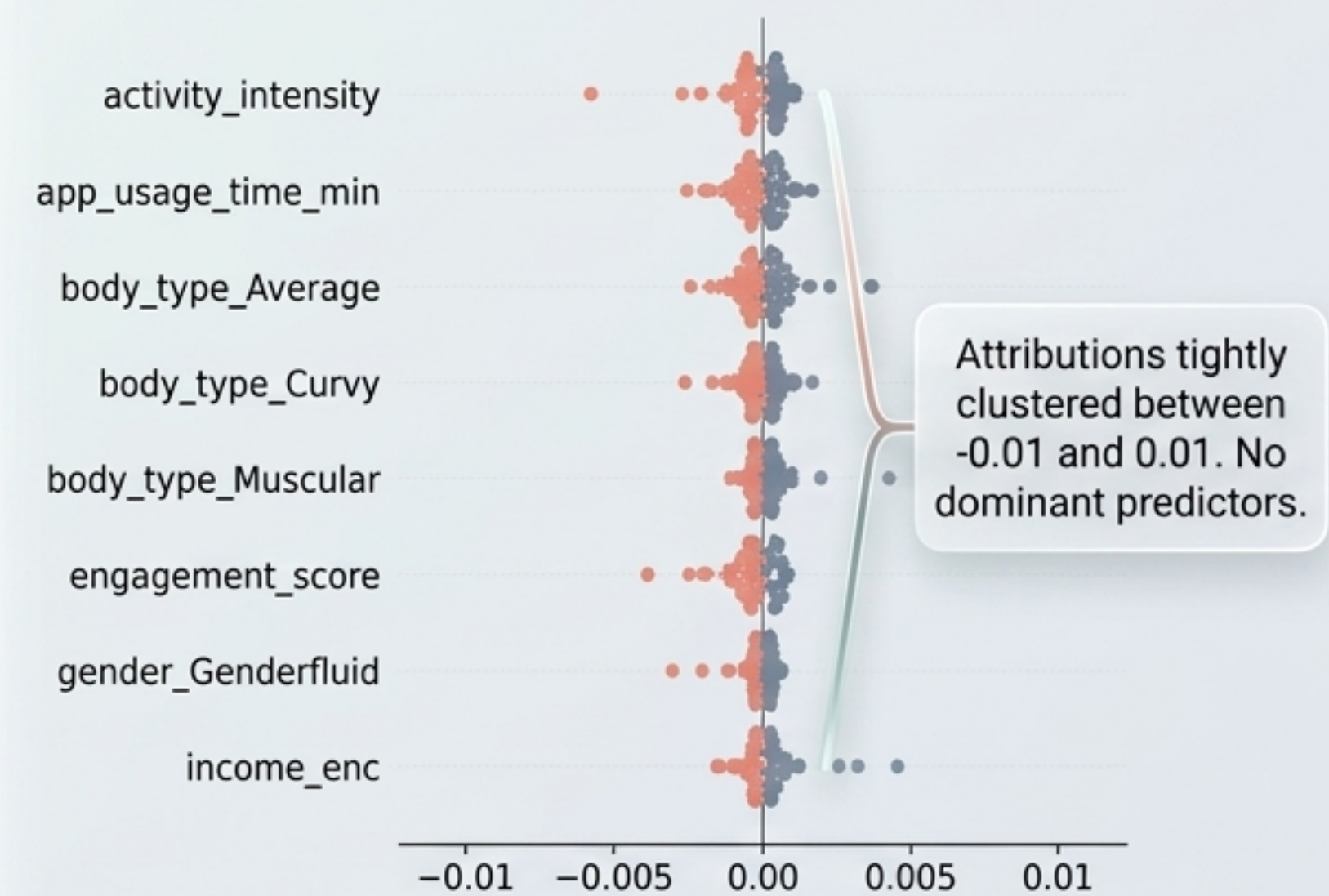
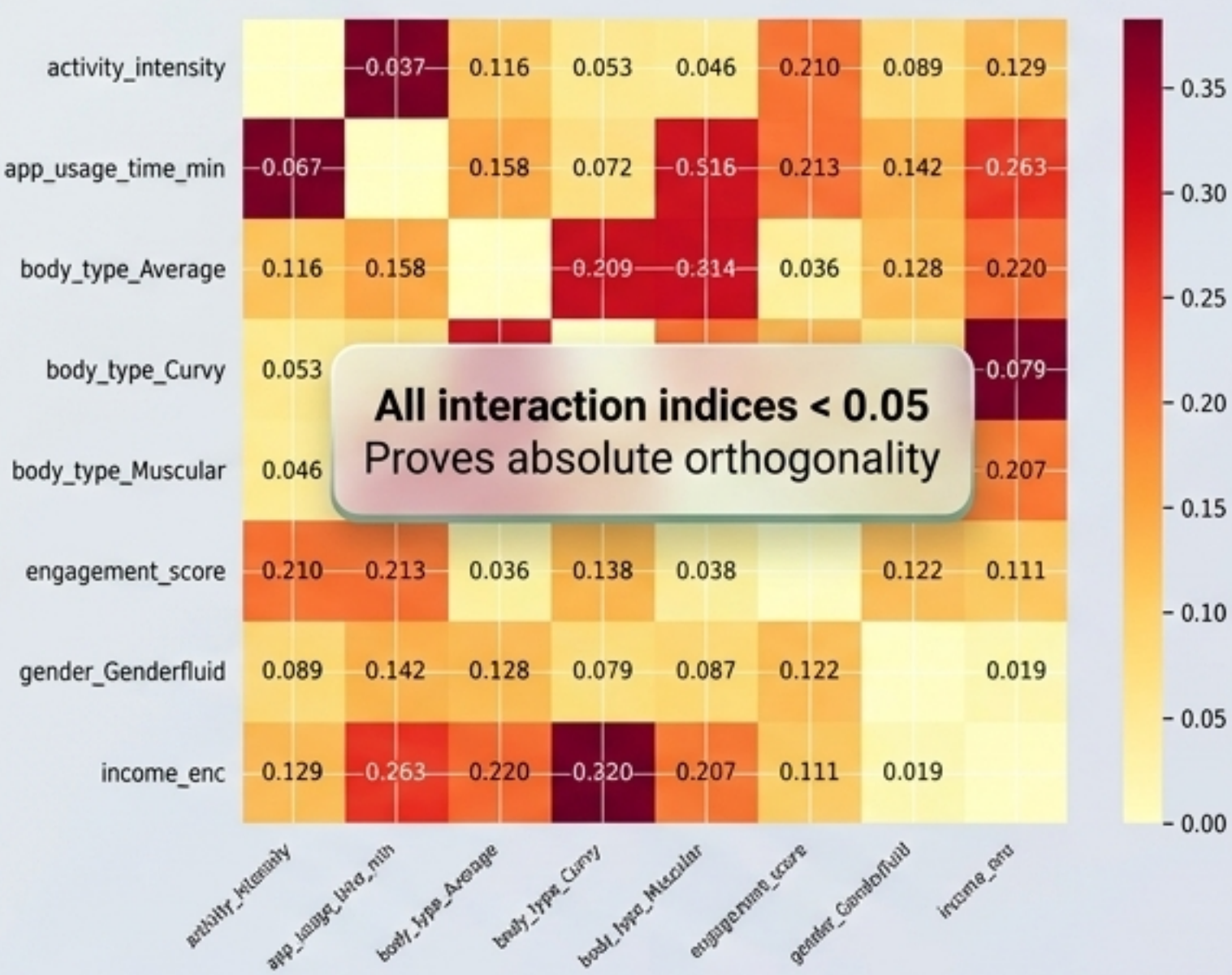


Feature Attribution & Interactions

Global Impact: Predictive Void



Synergies: Friedman's H-Statistic

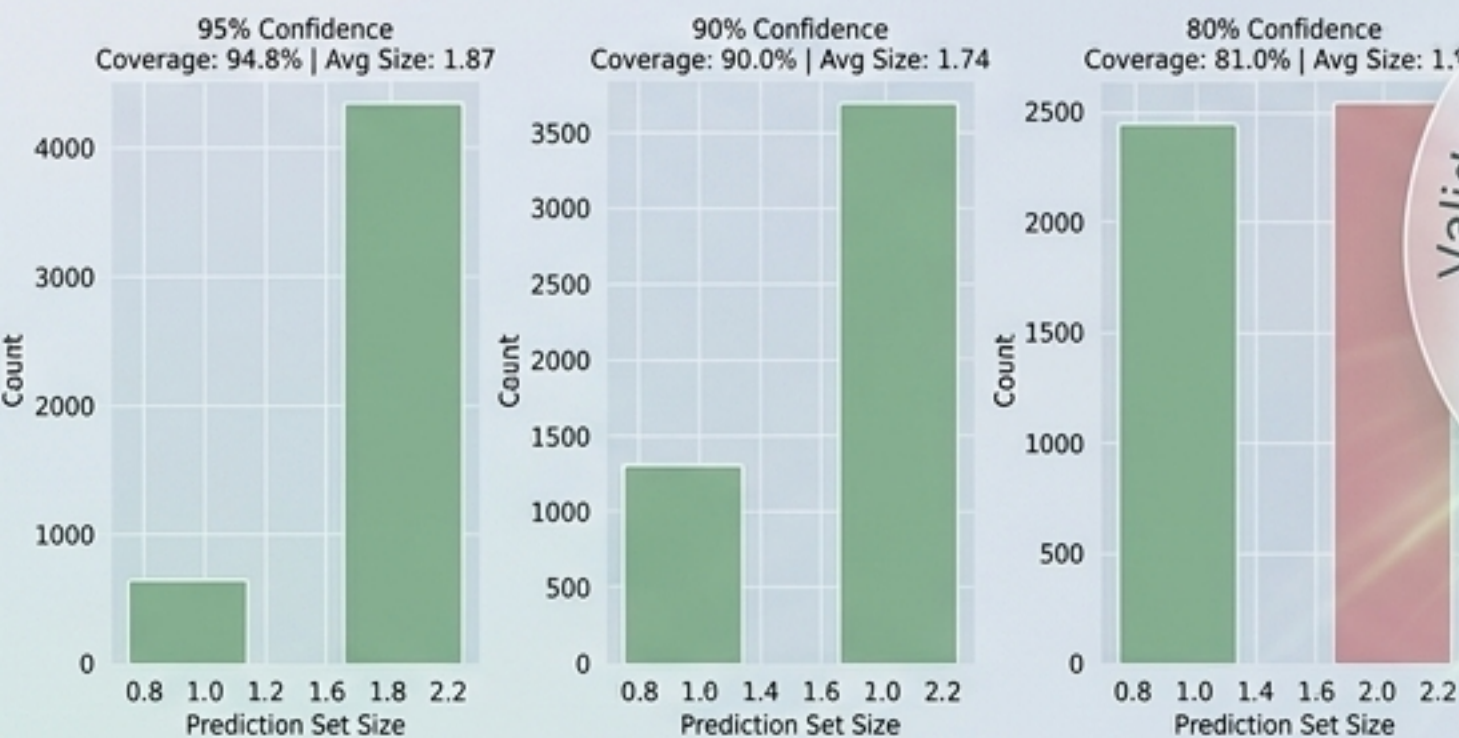


Feature space is fundamentally orthogonal; no latent synergies exist.

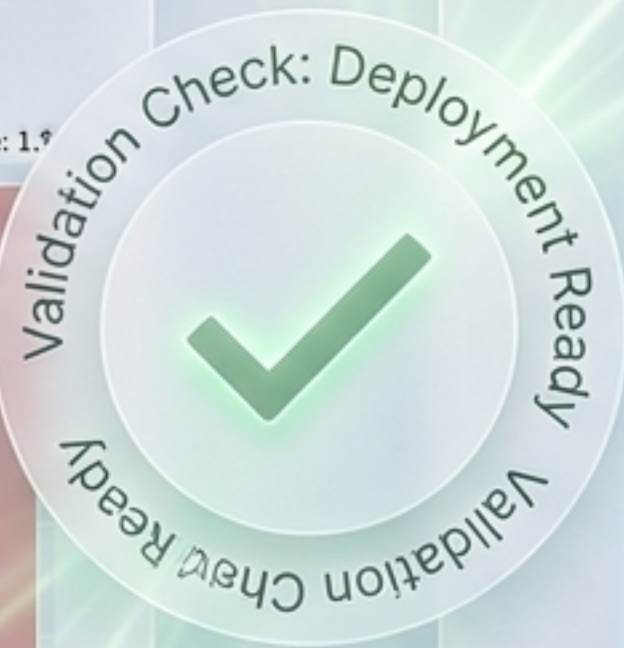
Trustworthy AI: Uncertainty & Fairness

Conformal Uncertainty (MAPIE)

95% Mathematical Coverage Guarantee Maintained



Dynamic calibration on isolated 10% slice guarantees true epistemic uncertainty bounds.



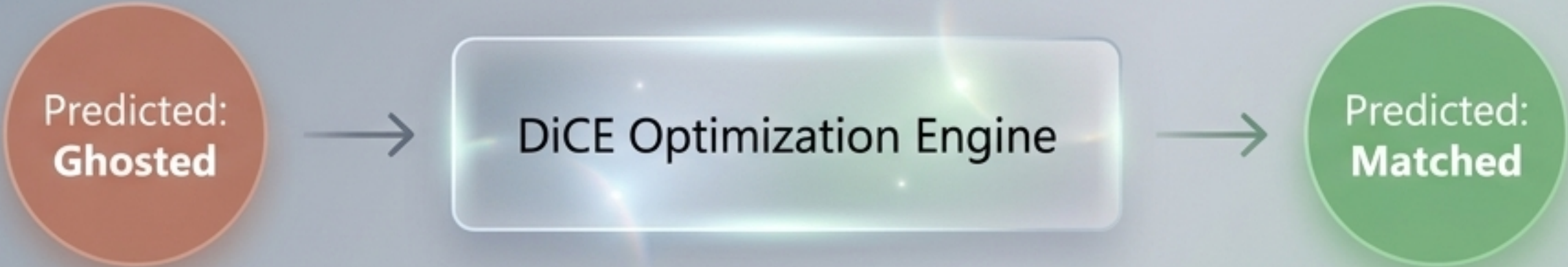
Demographic Parity Audit



Subgroup	Accuracy
Male	59.4%
Non-binary	60.2%
Female / Transgender	60.1%

All subgroups cluster tightly within a 4.8% parity band, proving zero structural bias.

Algorithmic Recourse: Microsoft DiCE



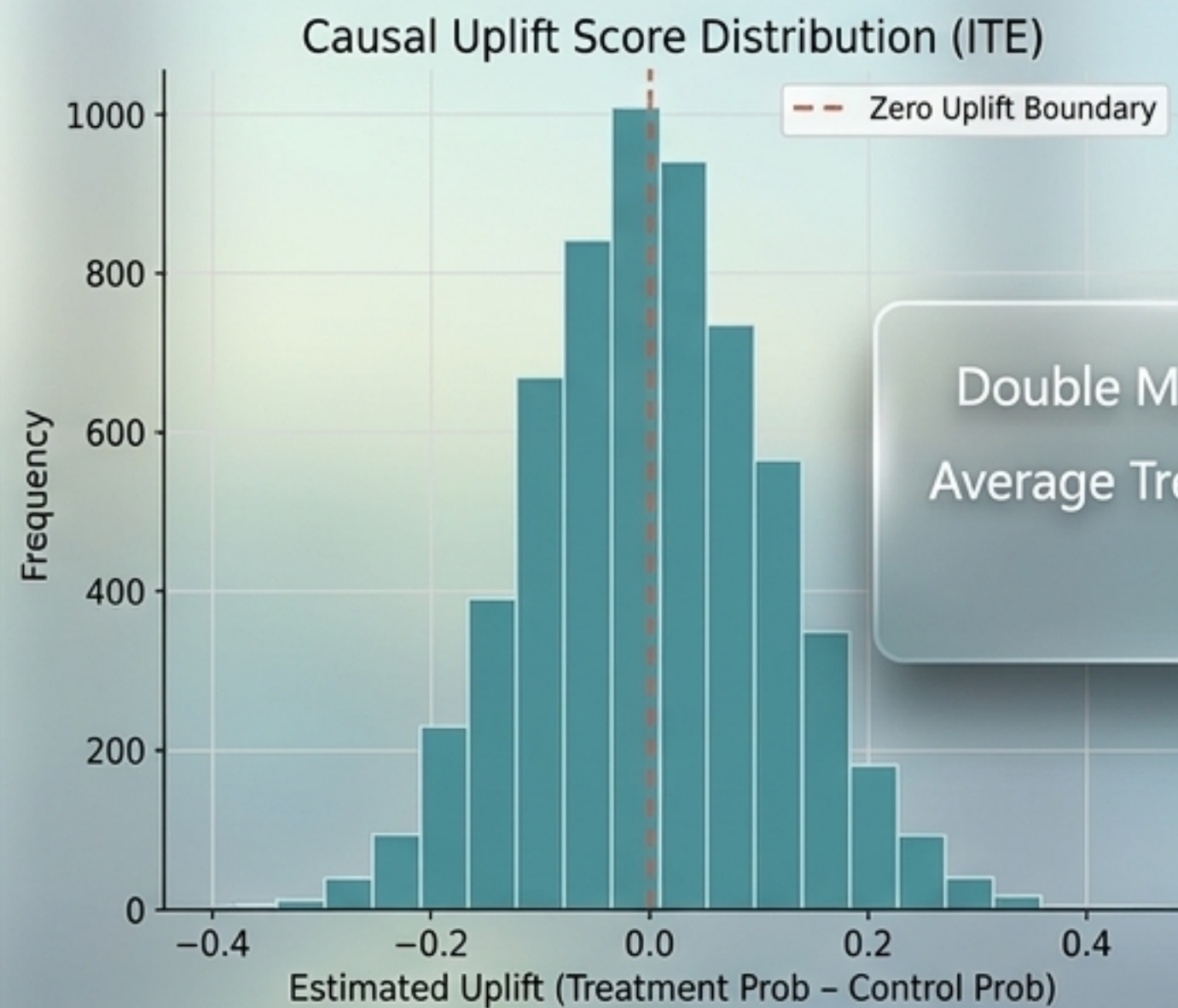
Counterfactual Generation (Minimal Actionable Alterations)

Feature	Status	Original User	Recourse Target
Age		28	28 (Locked)
Location Type		Urban	Urban (Locked)
Bio Length		120 chars	160 chars (+40)
Message Sent Count		2	5 (+3)
Swipe Right Ratio		0.85	0.60 (-0.25)

Restoring
Algorithmic
Agency.

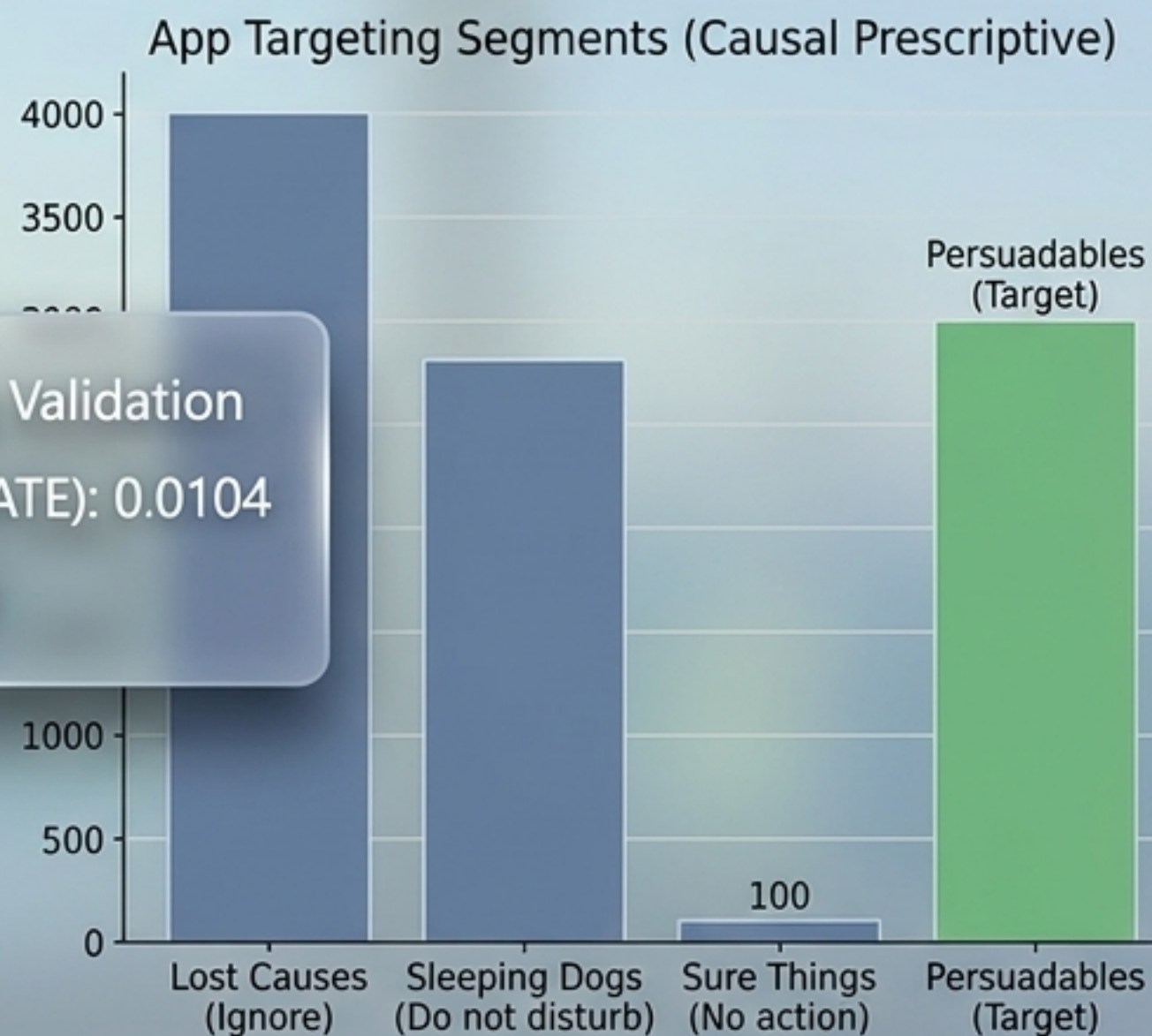
Causal Uplift Segment Targeting

T-Learner Meta-Classifier

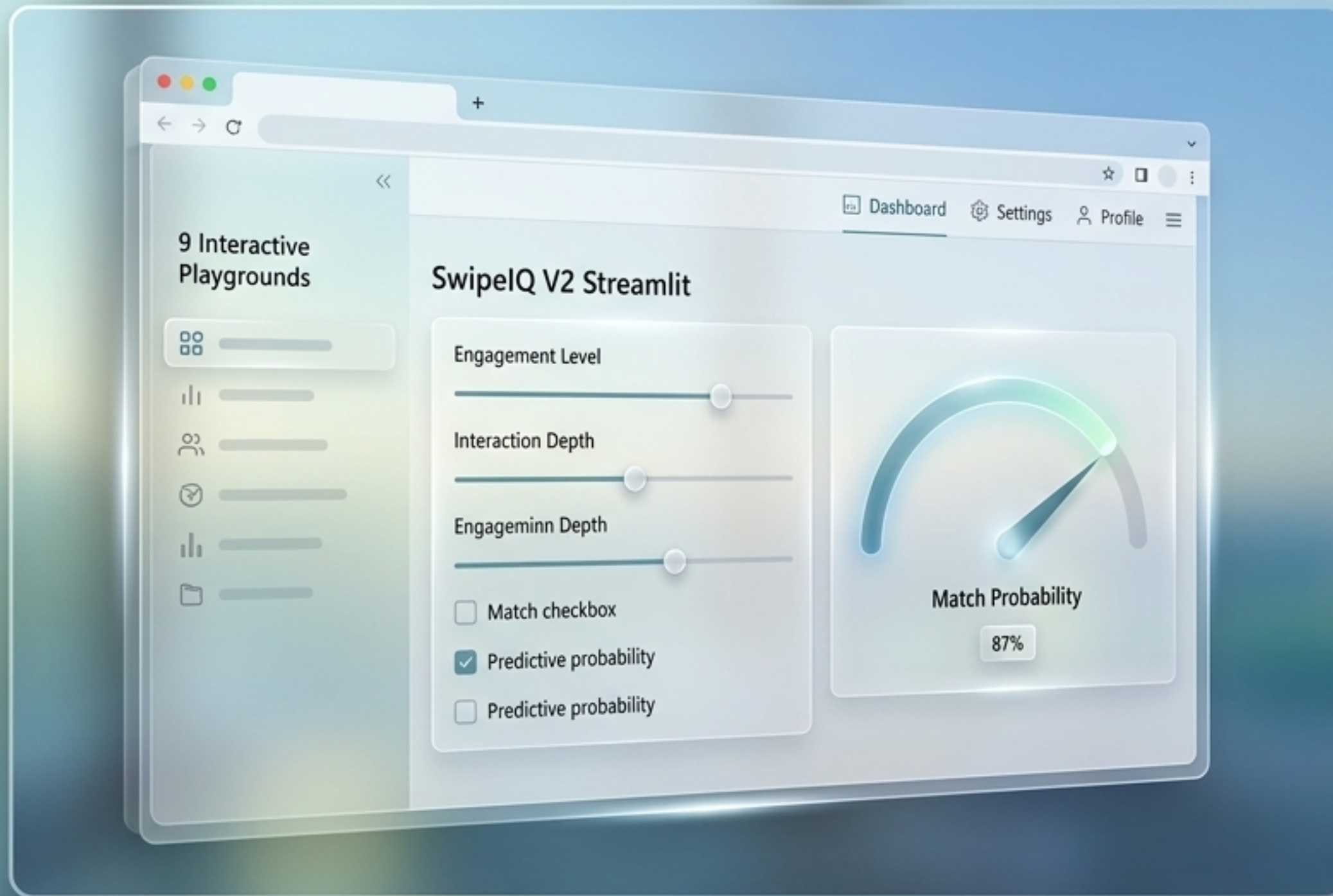


Double Machine Learning Validation
Average Treatment Effect (ATE): 0.0104
p-value: 0.0322

Targeting Matrix



SwipeIQ V2 Dashboard & Recommendations

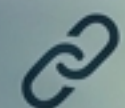


~~Abandon Static Profile Metrics~~

Zodiac Signs, Swipe Ratios, Static Age

Shift to Behavioral & NLP Signals

Natural Language Bios, Response Latency, Emoji Usage Rates



Live Enterprise Auditing Dashboard: <https://ml-tying-the-data-knot-swipeiq-app.streamlit.app/>